

Date 20/01/22/06
Class BSc 2nd Year
Roll No. 68
Shift Day

Physics Practical Sheet

Prithvi Narayan Campus

Experiment No. 01
Group
Sub Physics
Set

Object of the Experiment (Block Letters) : TO CONSTRUCT FUNDAMENTAL GATES
OR AND AND BY USING DIODE, TRANSISTOR AND LEAD.

APPARATUS REQUIRED:

- (i) Diode (ii) Transistor (iii) Lead
(iv) Bread board (v) Wires (vi) Multimeter

THEORY:

(I) OR GATE

The electronic symbol for a two input OR gate is shown in figure (a) and its equivalent circuit in fig (b). The two input have been marked as A and B and the output 'Y'. It is worth reminding the reader as per Boolean algebra, the three variables A, B and Y have only one of the two values i.e. either 0 or 1.

Truth Table

<u>Inputs</u>		<u>Output</u>
A	B	$Y = A + B$
0	0	0
0	1	1
1	0	1
1	1	1

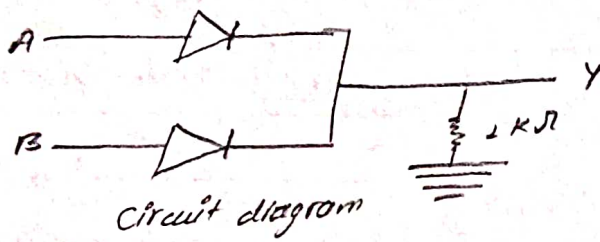
We may interpret the truth table as follows:

When both input are 0, output Y is 0 (Lamp is off).

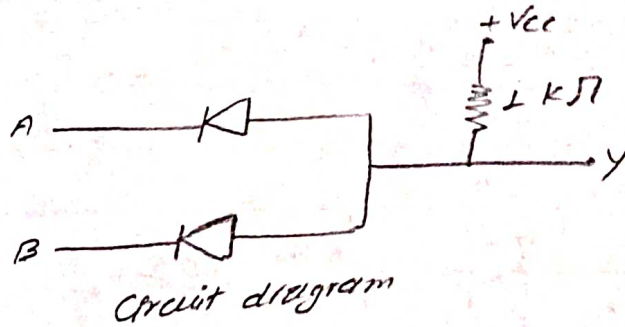
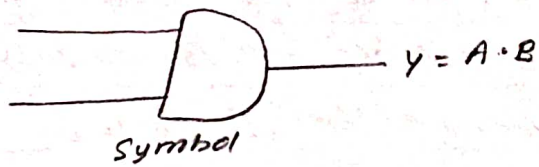
When A in logic states 0 but B in logic state 1 (switch 'B' is closed), the output Y is logic state 1 (Lamp is on).

Lamp would be also on when 'A' is closed and 'B' is open. Of course, lamp would be on when both switches are closed. It is so because an OR gate is equivalent to a parallel circuit in its logic function.

(I) OR GATE



(II) AND GATE



(II) AND GATE:

The electronic symbol of AND gate is shown in figure below and equivalent circuit in along side. It is worth reminding the readers once again that the three variables A, B, Y have a value of either 0 or 1.

Logic Operations:

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

- (i) The AND gate gives an output only when all its input are present.
- (ii) The AND gate has a 1 output when both A and B are 1. Hence this gate is an all-horunder or nothing gate whose output occurs only when all its input are perfect.
- (iii) In True/False terminology, the output of an AND gate will be true only if all input are true its output would be false if any of its input are false.

OBSERVATION:

(I) OR GATE:

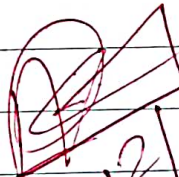
S.N	Inputs		Output
	A	B	$Y = A + B$
1.	0	0	0
2.	0	3.21	2.66
3.	3.21	0	2.58
4.	3.21	3.21	2.61

(II) AND GATE:

S.N	Inputs		Output
	A	B	$Y = A \cdot B$
1.	0	0	0.50
2.	0	3.21	0.56
3.	3.21	0	0.54
4.	3.21	3.21	3.93

RESULT:

From the above experiment, the function of simple logic gates are observed.


12/07